

CODE:-

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract BankAccount {
    // State variable to store balance of the account holder
    mapping(address => uint256) balances;
    // Deposit money into the account
    function deposit(uint256 amount) public payable {
        require(amount > 0, "Deposit must be greater than zero");
        balances[msg.sender] += amount;
    }
    // Withdraw money from the account
    function withdraw(uint256 amount) public {
        require(amount <= balances[msg.sender], "Insufficient balance");
        balances[msg.sender] -= amount;
    }
    // Show account balance
    function getBalance() public view returns (uint) {
        return balances[msg.sender];
    }
}
```

OUTPUT:-

The screenshot displays the Remix IDE interface. On the left, the 'SOLIDITY COMPILER' panel shows the compiler version '0.8.30+commit.73712a01' and a 'Compile bank.sol' button. The main editor area shows the Solidity code for the 'BankAccount' contract. The code is as follows:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract BankAccount {
5     // State variable to store balance of the account holder
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10         require(amount > 0, "Deposit must be greater than zero");
11         balances[msg.sender] += amount;
12     }
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17         balances[msg.sender] -= amount;
18     }
19
20     // Show account balance
21     function getBalance() public view returns (uint) {
22         return balances[msg.sender];
23     }
24 }
25
```

The bottom of the interface shows the 'ABI' and 'Bytecode' tabs, and a status bar at the very bottom with a 'Scam Alert' and a 'Did you know?' message.

REMIX v1.0.0

default_workspace

Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

ENVIRONMENT

Remix VM (Prague)

VM

ACCOUNT

0x583...eddC4 (99.9999999999)

Authorize Delegation

GAS LIMIT

Estimated Gas

Custom 3000000

VALUE

0 Wei

CONTRACT

BankAccount - banksol

evm version: prague

Deploy

Publish to IPFS

At Address Load contract from Address

Transactions recorded 0

Compiled

Home

banksol

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1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
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```

Explain contract

AI copilot

0 Listen on all transactions Filter with transaction hash or address

Scam Alert Initialize as git repo Did you know? To learn new contract patterns and prototype, you can activate and try the cookbook plugin! RemixAI Copilot (enabled)

REMIX v1.0.0

default_workspace

Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

CONTRACT

BankAccount - banksol

evm version: prague

Deploy

Publish to IPFS

At Address Load contract from Address

Transactions recorded 9

Deployed Contracts

BANKACCOUNT AT 0x907...B5

Balance: 0 ETH

deposit 1000

withdraw uint256 amount

getBalance

uint256: 1000

Low level interactions

CALLDATA

Transact

Compiled

Home

banksol

```
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```

Explain contract

AI copilot

0 Listen on all transactions Filter with transaction hash or address

creation of BankAccount pending...

[vm] from: 0x583...eddC4 to: BankAccount.(constructor) value: 0 wei data: 0x688...e8033 logs: 0 hash: 0x212...f2291 Debug

transact to BankAccount.deposit pending ...

[vm] from: 0x583...eddC4 to: BankAccount.deposit(uint256) 0x907...b5E99 value: 0 wei data: 0xb6b...003e8 logs: 0 hash: 0x631...9636a Debug

call to BankAccount.getBalance

[call] from: 0x58380a6a781c568545dcfc803fc8875f56beddC4 to: BankAccount.getBalance() data: 0x120...65fe0 Debug

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REMIXv1.0.0

default_workspace

Login with GitHubTheme

DEPLOY & RUN
TRANSACTIONS

CONTRACT

BankAccount - bank.sol

evm version: prague

Deploy

Publish to IPFS

At AddressLoad contract from Address

Transactions recorded10i

Deployed Contracts1

BANKACCOUNT AT 0x9D7...B5

Balance: 0 ETH

deposit1000

withdraw500

getBalance

uint256: 500

Low level interactions

CALLDATA

Transact

Compiled

Homebank.sol

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract BankAccount {
5     // State variable to store balance of the account holder
```

Explain contract

AI copilot

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creation of BankAccount pending...

[vm] from: 0x583...eddC4 to: BankAccount.(constructor) value: 0 wei data: 0x688...e8033 logs: 0 hash: 0x212...f2291

Debug

transact to BankAccount.deposit pending ...

[vm] from: 0x583...eddC4 to: BankAccount.deposit(uint256) 0x907...b5E99 value: 0 wei data: 0xb6b...003e0 logs: 0 hash: 0x631...9636a

Debug

call to BankAccount.getBalance

call [call] from: 0x58380a6a701c568545dCfcB03FcB875f56beddC4 to: BankAccount.getBalance() data: 0x120...65fe0

Debug

transact to BankAccount.withdraw pending ...

[vm] from: 0x583...eddC4 to: BankAccount.withdraw(uint256) 0x907...b5E99 value: 0 wei data: 0x2e1...001f4 logs: 0 hash: 0x3cf...73f49

Debug

call to BankAccount.getBalance

call [call] from: 0x58380a6a701c568545dCfcB03FcB875f56beddC4 to: BankAccount.getBalance() data: 0x120...65fe0

Debug

Scam AlertInitialize as git repo

Did you know? To learn new contract patterns and prototype, you can activate and try the cookbook plugin!

RemixAI Copilot (enabled)

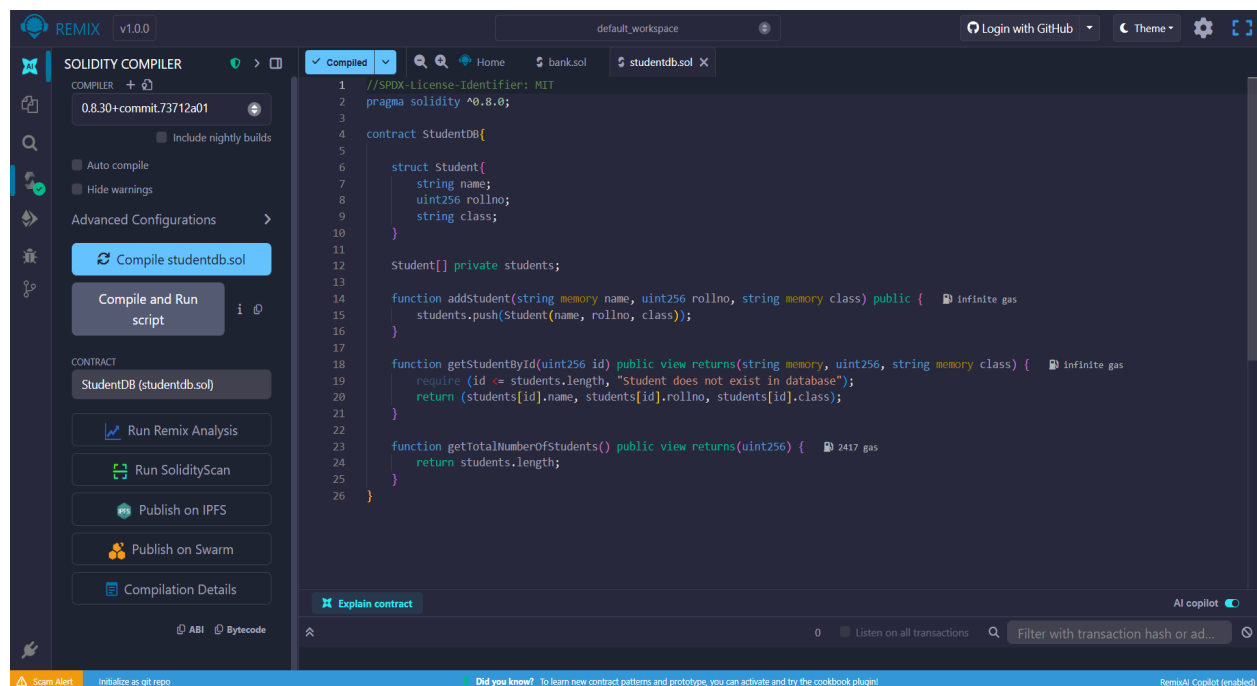
CODE:-

```
//SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract StudentDB{
    struct Student{
        string name;
        uint256 rollno;
        string class;
    }
    Student[] private students;
    function addStudent(string memory name, uint256 rollno, string memory class) public {
        students.push(Student(name, rollno, class));
    }

    function getStudentById(uint256 id) public view returns(string memory, uint256, string
memory class) {
        require (id <= students.length, "Student does not exist in database");
        return (students[id].name, students[id].rollno, students[id].class);
    }

    function getTotalNumberOfStudents() public view returns(uint256) {
        return students.length;
    }
}
```

OUTPUTS:-



REMIX v1.0.0

default_workspace

Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

CONTRACT

StudentDB - studentdb.sol

evm version: prague

Deploy

Publish to IPFS

At Address Load contract from Address

Transactions recorded 12 i

Deployed Contracts 1

STUDENTDB AT 0x0FC...9A836

Balance: 0 ETH

ADDSTUDENT

name: ABC

rollno: 21

class: BE

Calldata Parameters

transact

getStudentById uint256 id

Compiled

1 //SPDX-License-Identifier: MIT

2 pragma solidity ^0.8.0;

3

4 contract StudentDB{

5

Explain contract

AI copilot

0 Listen on all transactions Filter with transaction hash or ad...

creation of StudentDB pending...

[vm] from: 0x583...eddC4 to: StudentDB.(constructor) value: 0 wei data: 0x608...e0033 logs: 0 hash: 0xd4...99d57 Debug

transact to StudentDB.addStudent pending ...

[vm] from: 0x583...eddC4 to: StudentDB.addStudent(string,uint256,string) 0x0FC...9A836 value: 0 wei data: 0xd3...00000 logs: 0 hash: 0xb5d...2efe0 Debug

Scan Alert Initialize as git repo

Did you know? To learn new contract patterns and prototype, you can activate and try the cookbook plugin!

RemixAI Copilot (enabled)

REMIX v1.0.0

default_workspace

Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

Transactions recorded 12 i

Deployed Contracts 1

STUDENTDB AT 0x0FC...9A836

Balance: 0 ETH

ADDSTUDENT

name: ABC

rollno: 21

class: BE

Calldata Parameters

transact

getStudentById 0

0 string: ABC

1 uint256: 21

2 string: class BE

getTotalNum... 0

0 uint256: 1

Low level interactions

CALLDATA

Transact

Compiled

1 //SPDX-License-Identifier: MIT

2 pragma solidity ^0.8.0;

3

4 contract StudentDB{

5

Explain contract

AI copilot

0 Listen on all transactions Filter with transaction hash or ad...

creation of StudentDB pending...

[vm] from: 0x583...eddC4 to: StudentDB.(constructor) value: 0 wei data: 0x608...e0033 logs: 0 hash: 0xd4...99d57 Debug

transact to StudentDB.addStudent pending ...

[vm] from: 0x583...eddC4 to: StudentDB.addStudent(string,uint256,string) 0x0FC...9A836 value: 0 wei data: 0xd3...00000 logs: 0 hash: 0xb5d...2efe0 Debug

call to StudentDB.getStudentById

call [call] from: 0x58380a6a701c568545dcfc803fc8075f56beddC4 to: StudentDB.getStudentById(uint256) data: 0xc8f...00000 Debug

call to StudentDB.getTotalNumberOfStudents

call [call] from: 0x58380a6a701c568545dcfc803fc8075f56beddC4 to: StudentDB.getTotalNumberOfStudents() data: 0x398...4403a Debug

Scan Alert Initialize as git repo

Did you know? To learn new contract patterns and prototype, you can activate and try the cookbook plugin!

RemixAI Copilot (enabled)